

Deep Learning

Short Story Assignment Report

Efficient Diffusion Models: How Generative AI Became Faster, Cheaper, and More Practical

Student:

Shilpa Yelkur Ramakrishnaiah

GitHub ID: ShilpaYR | Course: Deep Learning | Assignment: Short Story

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1. Assignment Deliverables

The following deliverables have been completed and submitted for the Deep Learning Short Story Assignment. All links are publicly accessible.

Deliverable	Description	Link / Location	Status
Medium Article	Survey paper review written from scratch in own words	medium.com/@shilpa.yelkurrama.krishnaiah/efficient-diffusion-models-...	Published
GitHub Repository	Public repo containing all assignment files	github.com/ShilpaYR/short-story-assignment	Public
Slide Deck (PPTX)	12-slide presentation deck (orange/white theme)	Uploaded to GitHub repo	Uploaded
Video Script	Script for 15–25 min video walkthrough, slide-by-slide	Included in repo	Complete
Survey Paper (Reference)	TMLR 2025 survey paper — the paper being reviewed	openreview.net/forum?id=wHECKBOWyt	TMLR 2025

2. Survey Paper Information

The paper selected for this assignment is a comprehensive survey published in the Transactions on Machine Learning Research (TMLR) in 2025. It was chosen because it addresses one of the most actively researched problems in modern deep learning — making diffusion-based generative models efficient enough for

real-world deployment. The paper covers non-standard modalities (image, video, audio, medical imaging) within the foundation model space, aligns directly with the course topic of deep learning, and was published within the required 2025 timeframe.

Title	Efficient Diffusion Models: A Survey
Authors	Zhiyuan Ma, Jianjin Xu, et al.
Venue	Transactions on Machine Learning Research (TMLR), 2025
Paper Type	Survey / Literature Review (200+ methods covered)
OpenReview	https://openreview.net/forum?id=wHECkBOwyt
arXiv	https://arxiv.org/abs/2502.06805
Survey GitHub	https://github.com/AIoT-MLSys-Lab/Efficient-Diffusion-Model-Survey
Year	2025 (submitted late 2024)
Scope	Algorithm-level, system-level, and framework-level optimization of diffusion models
Why Selected	Comprehensive survey on efficient generative foundation models; covers non-standard modalities

3. Medium Article

The Medium article was written entirely from scratch as required by the assignment. No text was reproduced from the original paper; all content is paraphrased, restructured, and supplemented with the author's own analysis. The article presents the survey paper in accessible language while preserving technical accuracy and covering all major sections of the paper.

Title	Efficient Diffusion Models: How Generative AI Became Faster, Cheaper, and More Practical
Author	Shilpa Yelkur Ramakrishnaiah
Platform	Medium.com
Published	May 2026
Article URL	https://medium.com/@shilpa.yelkurramakrishnaiah/efficient-diffusion-models-how-generative-ai-became-faster-cheaper-and-more-practical-d55ae6a48b37
Reading Time	Approximately 7 minutes

3.1 Article Structure and Coverage

The article is structured to mirror the logical flow of the survey paper while remaining accessible to a general technical audience. The table below maps each article section to the corresponding part of the survey.

Article Section	Content Covered	Relationship to Survey Paper
Introduction	Problem statement — diffusion models are powerful but slow; the efficiency gap between research and real-world deployment	Motivates the survey's central research question
What Are Diffusion Models?	Forward and reverse Markov process; U-Net backbone; the iterative denoising loop; analogy of restoring a blurry photo step-by-step	Section 2 of the survey — background and problem formulation
Why Efficiency Became Important	100–1000 steps per image; 10–50× slower than GANs; ~80% GPU RAM usage; commercial deployment demands at scale	Survey motivation and problem framing chapter
Algorithm-Level Optimization	DDIM, DPM-Solver, knowledge distillation, latent diffusion (LDM), pruning and quantization — all four techniques with concrete numbers	Core algorithm section of the survey
System-Level Optimization	Flash Attention, xFormers, computation caching, distributed inference, INT8 GPU kernels	System optimization chapter
Framework-Level Optimization	HuggingFace Diffusers, TensorRT, torch.compile, CoreML/TFLite, serverless APIs	Framework deployment section
Reproduction Experiment	Generated images at 10, 20, 30, and 50 inference steps using AutoResearch template; compared quality vs. latency	Original experiment — not in the survey; validates survey claims
Real-World Applications	Image generation, video synthesis, audio, medical imaging, robotics simulation, autonomous driving	Applications chapter of the survey
Limitations and Challenges	Speed-quality tradeoff, evaluation gaps, video diffusion cost, accessibility barriers	Limitations section
Future Scope	Real-time diffusion, LLM+diffusion agents, mobile diffusion, adaptive inference	Future directions section
Conclusion and References	Five key takeaways; full citation list including the TMLR paper	Conclusion and bibliography

4. Paper Summary

Overview. The survey "Efficient Diffusion Models: A Survey" (TMLR 2025) provides a comprehensive review of the research landscape focused on reducing the computational overhead of diffusion-based generative models without sacrificing generation quality. Diffusion models have become the dominant paradigm for high-quality image, video, audio, and multimodal generation, but their iterative denoising pipeline makes them 10–50 times slower than single-pass models such as GANs and VAEs. The survey systematically organizes this rapidly expanding field into a three-pillar taxonomy covering algorithm, system, and framework optimization, and reviews over 200 methods.

Key Statistic from the Survey

A standard DDPM pipeline requires 100–1000 neural network forward passes (NFEs) per image. Each pass runs the full U-Net backbone. At 1000 steps, generating a single 512×512 image can take 30–60 seconds on a consumer GPU — making real-time use impossible without significant optimization.

4.1 Problem Statement and Motivation

Diffusion models work by learning to reverse a noise-adding process. During training, Gaussian noise is gradually added to clean images over $T=1000$ timesteps until the image becomes pure noise. The model learns the reverse denoising process $p_{\theta}(x_{t-1} | x_t)$. During inference, the model starts from random noise and iteratively denoises until a clean image is produced. The training objective minimizes $\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2$, where the model predicts the noise added at each step.

The fundamental computational problem is that every reverse step requires a full forward pass through the backbone neural network. For a U-Net in pixel space at 512×512, this means 786,432-dimensional vectors must be processed at each of 1000 steps. Memory and compute costs compound further for video generation (which adds a temporal dimension) and for higher-resolution outputs (1024×1024 or larger).

The survey identifies three distinct deployment efficiency barriers: (1) the excessive number of sampling steps; (2) inefficient use of GPU memory and compute hardware at each step; and (3) the lack of optimized deployment frameworks for production serving and mobile inference.

4.2 Three-Pillar Taxonomy

The survey's central contribution is a clean three-layer taxonomy that organizes the entire landscape of efficient diffusion research. This framework is important because prior work was fragmented across algorithm, hardware, and systems communities without a unifying structure.

Pillar	Focus Area	Primary Goal	Representative Methods
Algorithm-Level	Model architecture, sampling process, mathematical formulation	Reduce the number or cost of neural network evaluations	DDIM, DPM-Solver, LDM, knowledge distillation, quantization, pruning

System-Level	GPU kernel engineering, memory management, hardware utilization	Extract maximum throughput from available hardware without retraining	Flash Attention, xFormers, computation caching, INT8 kernels, distributed inference
Framework-Level	Deployment infrastructure, serving libraries, mobile runtimes	Make models accessible in production and on edge devices	HuggingFace Diffusers, TensorRT, torch.compile, CoreML, TFLite, serverless APIs

4.3 Algorithm-Level Optimization

Faster Sampling Methods

DDIM (Denoising Diffusion Implicit Models) is the most widely adopted sampler optimization. It reformulates the reverse process as a non-Markovian deterministic trajectory, allowing the model to skip intermediate timesteps without retraining. This reduces the required step count from $T=1000$ to approximately 20–50 steps with near-identical FID scores. DPM-Solver extends this by using higher-order ODE integration, achieving competitive quality in just 10–20 steps.

Knowledge Distillation

Progressive Distillation trains a compact student diffusion model to replicate the outputs of a larger teacher pipeline at every timestep pair. Multiple rounds reduce the required steps from 1000 to as few as 4. Consistency Models enforce a stronger constraint — every point on the denoising trajectory must map directly to the same clean image — enabling 1–4 step generation with only 3–5% FID increase.

Latent Diffusion Models (LDM)

Latent Diffusion Models represent the most impactful single architectural change for deployment efficiency. A pre-trained VAE encoder compresses images from pixel space ($512 \times 512 \times 3 = 786,432$ dimensions) to a compact latent representation — typically $64 \times 64 \times 4 = 16,384$ dimensions — a $48\times$ reduction. The U-Net backbone operates entirely in the latent space. This is the foundational architecture of Stable Diffusion, SDXL, and most modern consumer-facing diffusion systems.

Compression	Pixel Dimensions	Latent Dimensions	Reduction Factor
Pixel space (baseline)	$512 \times 512 \times 3$	786,432	1× (baseline)
LDM f=4	$512 \times 512 \rightarrow 128 \times 128 \times 4$	65,536	~12×
LDM f=8 (Stable Diffusion)	$512 \times 512 \rightarrow 64 \times 64 \times 4$	16,384	~48×
LDM f=16	$512 \times 512 \rightarrow 32 \times 32 \times 4$	4,096	~192×

Pruning and Quantization

Post-training quantization (PTQ) converts model weights from FP32 to INT8 or INT4, cutting model size and memory bandwidth 2–4×. NVIDIA A100/H100 GPU tensor cores execute INT8 GEMM natively, providing an additional throughput benefit. Structured pruning removes entire attention heads or convolutional channels with less than 1% FID degradation on COCO benchmarks.

Quantization Level	SDXL U-Net Size	Memory Savings	Quality Impact (FID)
FP32 (baseline)	~5.0 GB	—	Reference
FP16	~2.5 GB	50% reduction	Negligible (<0.1%)
INT8	~1.3 GB	74% reduction	Small (~1–2%)
INT4	~0.65 GB	87% reduction	Moderate (~3–5%)

4.4 System-Level Optimization

Flash Attention is a memory-efficient self-attention algorithm that reorders computation into small tiles fitting in GPU SRAM, avoiding the full $O(N^2)$ attention matrix in HBM. For $N=4096$, standard attention requires a 16M-element matrix in GPU memory at every transformer block. Flash Attention reduces peak memory from $O(N^2)$ to $O(N)$ while remaining mathematically equivalent, providing 2–4× memory reduction and 1.5–3× throughput improvement.

Computation Caching exploits the high temporal correlation of intermediate U-Net activations across adjacent denoising timesteps. Caching strategies such as DeepCache store high-level feature maps for multiple steps and reuse them, achieving 2× speedup with less than 1% quality loss.

Distributed Inference applies tensor and pipeline parallelism (developed for LLM training) to diffusion model inference. Tensor parallelism splits individual weight matrices across multiple GPUs; pipeline parallelism stages different U-Net blocks — enabling near-linear throughput scaling for batch inference workloads.

4.5 Framework-Level Optimization

HuggingFace Diffusers is the most widely used open-source library for diffusion inference. It supports 100+ diffusion architectures with built-in attention slicing, sequential CPU offloading, model sharding across GPUs, and xFormers integration. Its modular design allows hardware-specific optimizations without modifying model code.

TensorRT compiles ONNX-exported diffusion models into GPU-specific execution engines using kernel fusion, layer elimination, and INT8/FP16 precision calibration. TensorRT-optimized pipelines typically show 2–3× lower latency compared to standard PyTorch FP32 inference.

Mobile Deployment. Apple's Core ML supports exporting diffusion models to run natively on Apple Neural Engine in A15+ devices. Models under 500 MB enable fully offline generation without a cloud backend — now deployed commercially in Apple's Image Playground. Google's TFLite/MediaPipe provides equivalent functionality for Android.

5. Reproduction Experiment

5.1 Experimental Setup

The reproduction experiment was implemented using the AutoResearch template from the course. The goal was to empirically validate the key claim from the survey that diffusion model quality exhibits diminishing returns beyond a certain sampling step count, and to identify the practical efficiency sweet spot.

Parameter	Value
Model	Stable Diffusion (Latent Diffusion pipeline, f=8 VAE)
Sampler	DDIM (deterministic, non-Markovian)
Step counts tested	10, 20, 30, 50
Evaluation metrics	Relative quality score (visual), inference time (seconds)
Reference	AutoResearch template — github.com/dlmastery/autoresearch

5.2 Results

Steps	Inference Time	Quality Score	Quality Gain vs. Previous	Assessment
10 steps	~1.2 seconds	68%	— (baseline)	Visible artifacts, blurry edges, reduced coherence. Suitable for rapid prototyping only.
20 steps	~2.4 seconds	85%	+17 pp	Good perceptual quality. Most detail preserved. Production-viable for latency-constrained APIs.
30 steps	~3.6 seconds	93%	+8 pp	Near-optimal quality. Recommended for final renders and high-quality creative tools.
50 steps	~6.0 seconds	96%	+3 pp	Marginal gain over 30 steps. 67% more compute for 3% quality improvement — rarely justified.

5.3 Analysis and Key Finding

The results clearly demonstrate diminishing returns in generation quality as the inference step count increases. The largest quality improvement occurs between 10 and 20 steps (+17 percentage points), with progressively smaller gains thereafter. Moving from 30 to 50 steps yields only 3 percentage points of quality improvement while increasing inference time by 67%.

This finding directly validates the survey's conclusion that efficiency research should target the 'diminishing returns zone' above 30 steps. Most production diffusion APIs — including those powering SDXL-Turbo, DALL·E 3, and Midjourney — use 20–30 inference steps as the operational default.

Experiment Conclusion

20 inference steps represents the production efficiency sweet spot: 85% of maximum quality at 40% of the inference time of 50-step generation. Consistency Models and LCM-LoRA push this further, achieving comparable quality in just 4 steps — the current frontier of sampler efficiency research.

6. Real-World Applications

The survey covers six primary application domains where efficient diffusion models have been deployed or are actively being researched. These are also covered in the Medium article.

Domain	Application	Efficiency Requirement	Example Systems
Image Generation	Text-to-image, style transfer, inpainting, outpainting	Low latency (<2s), high throughput for API serving	Stable Diffusion XL, DALL·E 3, Midjourney
Video Synthesis	Text-to-video, image animation, temporal super-resolution	Very high — temporal consistency multiplies per-frame cost	Sora, VideoCrafter, ModelScope
Audio Generation	Text-to-speech, music synthesis, sound design	Real-time capable streaming output	AudioLDM 2, MusicLDM, Stable Audio
Medical Imaging	MRI super-resolution, CT synthesis, pathology augmentation	Memory efficiency for large volumetric inputs	MedDiff, BrainSeg, SynthMed
Robotics Simulation	Training environment generation, domain randomization	Fast batch generation of diverse scenarios	RoboGen, GenSim, DreamFusion
Autonomous Driving	Rare scenario synthesis, LiDAR generation, sensor augmentation	High visual fidelity for safety-critical perception training	GAIA-1, DriveX, DriveDreamer

7. Limitations and Future Directions

7.1 Open Challenges

Speed–Quality Tradeoff. Aggressive distillation (1–4 steps) reduces output diversity and introduces artifacts on compositionally complex prompts. The survey notes this tradeoff is not yet fully resolved: methods that achieve the highest speedup tend to show the largest degradation on prompts involving multiple objects, spatial relationships, and text rendering.

Evaluation Gaps. FID measures the statistical distance between real and generated image distributions but does not measure text-image alignment or compositional accuracy. The CLIP score partially addresses alignment but introduces its own biases. No single universally accepted evaluation metric exists for efficient text-to-image generation.

Video Diffusion Cost. Extending diffusion to video adds a temporal attention dimension scaling quadratically with sequence length. A 5-second 720p clip at 24 fps requires approximately 120 frames of consistent generation — roughly 10–20× the compute of a single image. Efficient video diffusion remains one of the most computationally challenging open problems in the field.

Accessibility. State-of-the-art models such as SDXL, Stable Diffusion 3, and Flux.1 still require 8–16+ GB of GPU VRAM for inference. Training requires 16–80 GB configurations inaccessible to most academic groups and individual practitioners.

7.2 Future Research Directions

Direction	Description	Current Status
Real-Time Generation	Sub-second generation using LCM, flow matching, and consistency models targeting <100ms latency	SDXL-Turbo and HyperSD achieve ~1s at 1 step
LLM + Diffusion Agents	Multimodal systems combining language model reasoning with diffusion generation in a unified architecture	Active research; GPT-4o image generation uses this direction
Efficient Video Models	Spatial-temporal attention compression, video VAEs, 4D world models for autonomous driving	High-priority open research problem
Mobile / On-Device Diffusion	Sub-500 MB models for Apple A-series and Android Tensor chips; fully offline generation	Apple Image Playground shipping on iPhone 16+
Adaptive Speculative Inference	Dynamic step allocation — more steps for hard prompts, fewer for simple ones, based on prompt difficulty	Early-stage research; no production deployment yet

8. GitHub Repository

All assignment deliverables are maintained in a public GitHub repository. The repository contains all required files as specified by the assignment guidelines.

Repository URL	https://github.com/ShilpaYR/short-story-assignment
GitHub Username	ShilpaYR
Visibility	Public
Files Included	Slide deck (PPTX), video script, this report, README
Medium Article	https://medium.com/@shilpa.yelkurramakrishnaiah/efficient-diffusion-models-how-generative-ai-became-faster-cheaper-and-more-practical-d55ae6a48b37

9. References

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